Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

1. Why does node F remain uninfected in the five-iteration SIR model simulation? 1 / 1 point

- It had too low a removal probability
- It was immune from the beginning
- No infected node pointed to it during the simulation
- It had already been removed

Nice work

Feedback: Node F had no incoming edge from an infected node; thus, it couldn't be infected.

2. What happens when a node enters the **removed class** in the SIR model? 1 / 1 point

- It becomes a seed node in future iterations
- It can still infect susceptible neighbors
- It transitions back to the infected class after a delay
- It is no longer able to infect or be infected

Nice work

Feedback: Removal signifies permanent immunity and exclusion from further diffusion dynamics.

3. According to the SIS model dynamics, what happens to a newly infected node? 1 / 1 point

- It infects all nodes in the network
- It tries to infect its neighbors and then becomes susceptible
- It immediately becomes susceptible again
- It remains infected for at least one full iteration

Nice work

Feedback: Newly infected nodes must wait until the following iteration before possibly becoming susceptible.

4. What is the **main difference** between the SI and SIS models? 1 / 1 point

- In SI, nodes can skip the infection state
- In SIS, infected nodes can return to being susceptible
- SI allows recovery while SIS does not
- SIS does not allow for infection spread

Nice work

Feedback: The key change in SIS is the ability of infected nodes to recover and return to the susceptible state.

5. What is the **primary purpose** of using the SI model in diffusion studies? 1 / 1 point

- To represent behavior adoption with feedback loops
- To account for recovery rates in epidemics
- To simulate decision-making in social networks
- To introduce the basic principles and assumptions of diffusion modeling

Nice work

Feedback: The SI model is a simple starting point for understanding diffusion, before exploring more complex models.

6. Which of the following best describes the **cascade size** in an SIS simulation? 1 / 1 point

- It is unaffected by the value of parameters P and S
- It always grows until all nodes are infected
- It can fluctuate over time as nodes become infected and recover
- It steadily decreases with each iteration

Nice work

Feedback: Cascade size in SIS can increase and decrease depending on infection and recovery rates.

7. If the value of parameter **S** is increased significantly while keeping **P** constant, what is the expected result? 1 / 1 point

- The number of infected nodes will likely decrease faster
- The network will split into disconnected components
- Infected nodes will become immune
- The infection will spread more rapidly

Nice work

Feedback: More nodes become susceptible more quickly, reducing infection persistence.

8. Why can the SI model eventually infect all nodes reachable from the seed node? 1 / 1 point

- Nodes develop new connections dynamically
- The infection rate increases over time
- Infected nodes never recover and keep attempting to infect neighbors
- All nodes eventually become seed nodes

Nice work

Feedback: Since infection is permanent, all reachable susceptible nodes will eventually be infected.

9. Which of the following statements are **true** about SIS model dynamics? 1 / 1 point

- [ ] Nodes eventually become permanently immune
- [ ] Recovery happens instantly after infection
- [x] Nodes can become infected multiple times

Nice work

Feedback: Since recovered nodes return to the susceptible state, reinfection is possible.

- [x] It involves two model parameters that control infection and recovery

Nice work

Feedback: The model uses 'p' for infection and 's' for recovery probabilities.

10. In the SI model, what happens to a node once it becomes infected? 1 / 1 point

- It returns to a susceptible state after one iteration
- It stays infected permanently and continues to infect susceptible neighbors
- It can no longer interact with other nodes
- It infects others only once and then becomes inactive

Nice work

Feedback: The SI model assumes perpetual infection and continuous attempts to spread the idea or virus.

11. Which of the following statements best describes how diffusion of innovations is similar to virus diffusion in networks? 1 / 1 point

- [ ] People are forced to adopt innovations from their neighbors
- [x] Ideas can "infect" people through social influence

Nice work

Feedback: This metaphor is used in the lecture to compare idea spread to virus transmission.

- [x] Individuals consider their neighbors' outcomes before deciding

Nice work

Feedback: Individuals observe their neighbors' experiences before choosing to adopt.

- [ ] All individuals in a network adopt innovations at the same time

12. Which parameter controls the **probability of removal** in the SIR model? 1 / 1 point

- r
- s
- p
- i

Nice work

Feedback: In the SIR model, 'r' is the probability that an infected node transitions to the removed class.

13. What is the role of the **second parameter 's'** in the SIS model? 1 / 1 point

- It determines the probability that an infected node becomes susceptible again
- It controls the number of seed nodes
- It represents the probability of a node becoming infected
- It dictates the network structure

Nice work

Feedback: The 's' parameter governs the recovery likelihood at each iteration.

14. Which of the following characteristics are true of the SI model? 1 / 1 point

- [x] It assumes permanent infection for each node

Nice work

Feedback: The core assumption of the SI model is that once infected, nodes stay infected.

- [ ] Nodes can return to a susceptible state after being infected
- [x] The model is considered the simplest form of the diffusion model

Nice work

Feedback: The lecture introduces SI as the most basic diffusion model.

- [ ] Infected nodes have a limited number of infection attempts

15. Which factor does **not** directly influence the diffusion of ideas or viruses in a population, according to the lecture? 1 / 1 point

- Network structure of the population
- Infectiousness of the virus or idea
- Time of day the diffusion starts
- Quality of the idea

Nice work

Feedback: The lecture does not mention time-of-day effects, focusing instead on idea quality, infectiousness, and network structure.

16. Which of the following statements are **true** about how cascade size changes in the SIR model? 1 / 1 point

- [ ] The cascade size is unaffected by r
- [x] A low r value can lead to most or all nodes becoming infected

Nice work

Feedback: With low removal probability, infected nodes can infect more before being removed.

- [ ] Cascade size increases indefinitely with a high r value
- [x] A high r value can prevent widespread infection

Nice work

Feedback: High removal probability can halt the cascade early.

17. Which of the following conditions are **likely to reduce** the spread of infection in the SIS model? 1 / 1 point

- [ ] Decreasing the value of s
- [x] Decreasing the value of p

Nice work

Feedback: Lower p reduces the likelihood of infection spread to neighbors.

- [ ] Increasing the number of seed nodes
- [x] Increasing the value of s

Nice work

Feedback: A higher s causes faster recovery, reducing the number of infected nodes.